Perimeter Magic Triangles



In 1972, Terrel Trotter, Jr., then a math teacher in Urbana Illinois, published an article called Normal Magic Triangles of Order *n* [1]. In 1974, he published a follow up article called Perimeter-magic Polygons [2]. In it he introduced the term ***Perimeter-magic Polygon***, and generalized it to include other polygonal shapes. Since that tim other writers have expanded on the same theme.

On this page, I will attempt to expand on, and summarize the published material on perimeter-magic polygons (PMPs for short) and to add interest, I will include some problems dealing with the subject matter. Because of the nature of perimeter-magic polygons, these problems will be suitable for middle school students. A companion page will feature magic graphs. That page also will include problems. However, these will tend to be much more complicated and will, in many cases, require the use o personal computer.

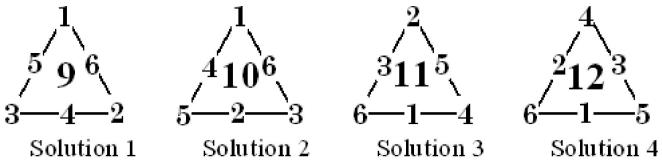
Order 3 Perimeter Magic Triangles	Some Math	Order 4 Perimeter Magic Triangles
Larger Perimeter Magic Triangles	Other Types of PM Triangles	Perimeter Magic Platonic Solids
	References	

Additional pages on this subject

PMPs with sides >3	Magic 3-D polygons and Graphs	Perimeter Magic cubes (on a previously written page)

Order 3 Perimeter Magic Triangles

Order-3 is the simplest perimeter- magic triangle, except for order 1 which is trivial (there are no order 2 perimeter-magic objects). Unlike magic squares, cubes, etc., t are multiple magic constants for the same order.



Here I show the only PMPs of order 3.

Solution 4 is the compliment of solution 1, and solution 3 is the complement of solution 2 (see below).

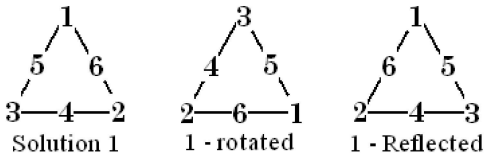
Complements.

For any given solution, there is always a complement solution. It is obtained by subtracting each number in turn from the sum of the first and last numbers in the series if your solution uses the numbers from 1 to 9, another solution is obtained by subtracting each number from 10 (ie. The sum of the first and last numbers in the series). feature applies to magic squares, cubes etc. as well.

The new solution obtained in this way must then be *normalized* by rotation and/or reflection, to comply with the above

For higher orders, with many solutions, it is necessary to only find solutions for the smallest half of the possible sums. The higher magic sums may then be obtained si by complementing these lower solutions.

Rotations and Reflections



Rotations and Reflections are not considered unique solutions. Each basic solution has two rotations and each of these has one (horizontal) reflection, for a total of 6 apparently different solutions. For other polygons, these numbers would obviously vary.

While any one of these six could be considered the basic solution, I have chosen to use the convention described in the next section.

Standard for Normalized solutions

There are many ways to sort magic perimeter solutions.
Other writers have chosen to sort by side or vertex totals (or possibly no sort at all).
For polygons with few solutions, this step is not important. However, as the number of solutions increases, a sorted list is necessary in order to prevent possible duplications.

I have chosen to use the following system.
To facilitate putting polygon solutions in order:
Place the lowest vertex number at the top
Then move clockwise around the polygon and
Enter vertex numbers in order, then
Enter side numbers in order
Arrange list of solutions in sorted order

Order 3 solutions in list order:

In the *sorted* list below, S=Side sum (i.e. the magic constant, V=Vertex sum, C=the complement solution #, Bold columns are vertices

#	A		B		C		S	V	C
1.	1	6	2	4	3	5	9	6	4
2.	1	6	3	2	5	4	10	9	3
3.	2	5	4	1	6	3	11	12	2
4.	4	3	5	1	6	2	12	15	1

Note that solutions 3 and 4 have been normalized before being added to the list
An example of complementing and normalizing:

#	A		B		C		S	V	C	
1.	1	6	2	4	3	5	9	6	4	Solution 1
4.	6	1	5	3	4	2				Subtract above number from 7
4.	4	3	5	1	6	2	12	15	1	Normalized solution 4

Some Math

Formulas required for solving perimeter-magic polygons.
Trotter covered the mathematic involved in great detail. Here I am only summarizing.

We are assuming the numbers used are consecutive starting with 1.
The following notation will be used.
k = the number of sides of the polygon
n = the number of integers per side (the order)
S = the magic constant (Trotter used C, but S is consistent with magic square terminology)
N = the number of consecutive integers being used
Vertex sum = k[S-1/2(n-1){(n-1)k+1}]

The smallest vertex sum is 1 + 2 + 3 i.e. the 3 smallest integers in the series from 1 to N.
The largest vertex sum is 3n – 3 (that is, the 3 largest integers).
However, because of parity, the 3 smallest integers and the 3 largest integers may not always give the possible smallest and largest vertex sums. There are two separate cases, depending on the values of n and k:

If n is even or n and k are both odd, then
$\min V_s = ½k(k + 1)$ $\max V_s = ½k[(2n – 3)k + 1]$
When n is odd and k is even, then
$\min V_s = ½k(k + 2)$ $\max V_s = ½(2n\text{-}3)k^2$

The final values we need to determine are the minimum and maximum values of the magic constant S.
Here again we must have two formulas, depending on the values of n and k.

If n is even or n and k are both odd, then
$\min S = (½n^2 – n + 1)k + ½ n)$ $\max S = (½n^2 – 1)k + ½ n$
When n is odd and k is even, then
$\min S = (½n^2 – n + 1)k + ½ (n + 1)$ $\max S = (½n^2 – 1)k + ½ (n – 1)$

A word of caution. These formulas indicate where solutions *may* be found. On very rare occasions, there may be no solution for a particular value of S. However, only such cases are known. They are; for a 4th order triangle with S = 18 or 22, and, 3rd order pentagon with S = 15 or 18.

For perimeter magic triangles, a simple equation gives the number of magic sums for any given order.
Number of different sums = 3n-5. This even allows for the special 'case of the two missing sums in the order 4 triangle

Can you find any other cases with no solution for a value of S between the theoretical minimum and maximum limits?
If there are any such cases, they will always be an even number because of the complement feature mentioned above.

A table generated from the above formulae has been placed at the beginning of each polygon section following.

[1] Terrel Trotter, Jr., *Normal Magic Triangles of Order n*, Journal of Recreational Mathematics, Vol. 5,, No. 1, 1972, pp.28-32
[2] Terrel Trotter, Jr., Perimeter-magic Polygons, Journal of Recreational Mathematics, Vol. 7,, No. 1, 1974, pp.14-20

A guide to finding solutions

I find the simplest method for constructing PMP's manually is the following:

k = the number of sides of the polygon
n = the number of integers per side (the order)
Vs = the sum of the vertex numbers
Ns = sum of the series of numbers used
Ms = the average mid-side sum (ie sum of integers between the two vertices of each side.

1. First choose the integers for the vertices from the set N where $N = (n\text{-}1)k$
2. The magic sum S where $S = (Ns+Vs)/n$ (this must be an integer if a valid solution is possible).
3. Find the average mid-side sum where $Ms = (Ns\text{-}Vs)/n$
4. Using the above step as a guide, partition the remaining numbers in the series into n groups with the required sums to place on the sides of the polygon (this shou always be possible if the result of step 2 was an integer.

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Order 4 Perimeter Magic Triangles

There are 18 basic solutions for order 4 PM Triangles. Note that there are no solutions with magic sums of 18 and 22.
The number of solutions for each sum should be symmetrical.

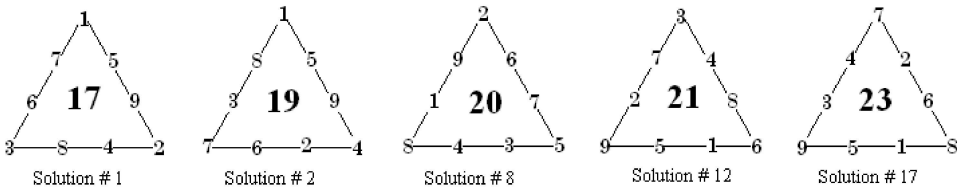
Sum	17	19	20	21	23
# of solutions	2	4	6	4	2 = 18

The list below contains all solutions for order 4 PM Triangles.

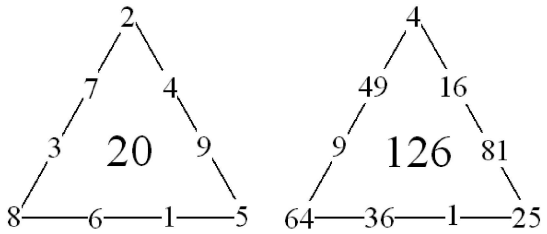
S=Side sum (i.e. the magic constant, V=Vertex sum, C=the complement solution #,
CS=the complement side sum, CV=the complement vertex sum. Columns A, B, and C are the vertices

#	A			B			C			S	V	C
1.	1	5	9	2	4	8	3	6	7	17	6	17
2.	1	5	9	4	2	6	7	3	8	19	12	12
3.	1	6	8	2	5	7	3	4	9	17	6	18
4.	1	6	8	4	3	5	7	2	9	19	12	14
5.	1	6	8	5	2	4	9	7	3	20	15	6
6.	2	4	9	5	1	6	8	3	7	20	15	5
7.	2	5	9	3	1	8	7	4	6	19	12	10
8.	2	6	7	5	3	4	8	1	9	20	15	8
9.	2	6	8	3	4	5	7	1	9	19	12	13
10.	3	2	9	7	1	5	8	4	6	21	18	7
11.	3	4	8	5	2	6	7	1	9	20	15	11
12.	3	4	8	6	1	5	9	2	7	21	18	2
13.	3	5	6	7	2	4	8	1	9	21	18	9
14.	3	5	7	6	2	4	9	1	8	21	18	4
15.	4	2	9	5	1	8	6	3	7	20	15	15
16.	4	3	8	5	2	7	6	1	9	20	15	16
17.	7	2	6	8	1	5	9	3	4	23	24	1
18.	7	3	5	8	2	4	9	1	6	23	24	3

Some example diagrams:



Solution # 17 is the normalized complement of solution #1, and # 12 is the normalized complement of #2.
Number 8 is it's own complement.



Solution number 6 is of special interest because it is bimagic. It was originally discovered by David Collison of California, USA [3]. When each number is squared, the triangle is still perimeter magic. This time with a sum of 126.

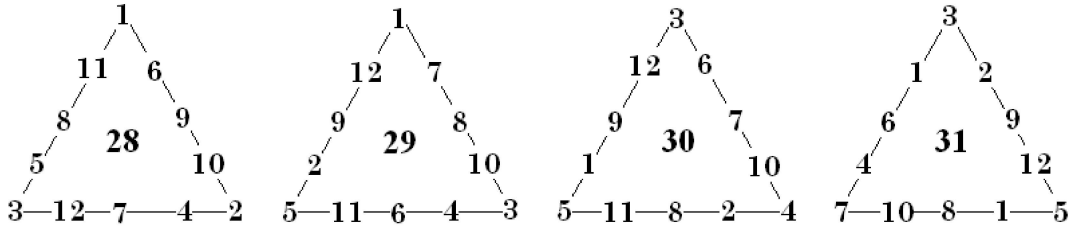
This suggest another investigation…to find bimagic solutions in other orders (or even trimagic solutions?).

[3] John R. Hendricks, *The Magic Square Course*, 2nd edition, 1992, 522 pages +,

Larger Perimeter Magic Triangles

Order-5 Triangles

Order 5 PM Traingles have ten values for S. They range from 28 to 37. Trotter [1] reports that there are 1356 basic (he calls them *primitive*) solutions.



Here I show solutions for the 4 lowest possible magic sums. By complementing these, you can obtain another solutions for the 4 highest sums. I leave it to you to find solutions for sums of 32 and 33.
Note that I have not supplied solution numbers to these. The only way that can be done is to find and list in order all of the possible basic solutions.

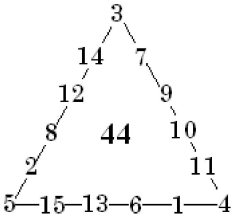
Orders-6 to 8 Triangles

Here is summary information on orders 6 to 8 perimeter magic triangles. For completeness, I included the same information for the smaller triangles.

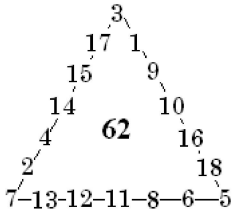
	Order 3	Order 4	Order 5	Order 6	Order 7	Order 8
Minimum S	9	17	28	42	59	79
Maximum S	12	23 *	37	64	74	97
Integers used	1 to 6	1 to 9	1 to 12	1 to 15	1 to 18	1 to 21
Minimum Vertex sum	6	6	6	6	6	6
Maximum Vertex sum	15	24	33	42	51	60
Number of basic solutions	4	18	1356?	?	?	?

* For order 4, there are no solutions with sums of 18 or 22.

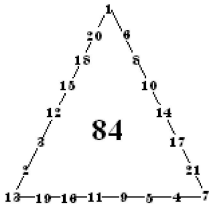
Following are sample solutions of orders 6, 7, and 8 PM Triangles. The reader is left with a challenge to find additional solutions. Or even, all the solutions for a giver order.



Order 6



Order 7



Order 8

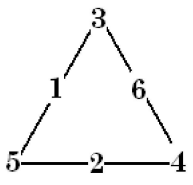
Perimeter magic triagonals are neither as complex, or mathematically challenging as magic squares or cubes.
Nevertheless, they can be of interest to anyone who is intrigued with the vast variety of number patterns.

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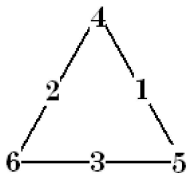
Other Types of PM Triangles

Subtractive magic triangles

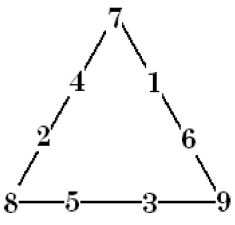
Sunday Ajose [4] proposed a type of triangle where the smallest number on each side is subtracted from the sum of the two largest numbers. For orders greater then 3, t sum of the smaller numbers is subtracted from the sum of the larger numbers on each side. In all cases, to be magic the differences must be the same. He calls this cons *d.* Some examples follow.



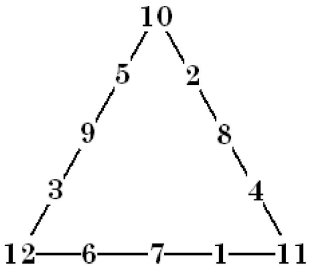
(4+6)-3=7
(4+5)-2=7
(3+5)-1=7



(4+5)-1=8
(5+6)-3=8
(6+4)-2=8



(7+9)-(1+6)=9
(9+8)-(3+5)=9
(8+7)-(2+4)=9



(10+8+11)-(2+4)=23
(11+7+12)-(1+6)=23
(12+9+10)-(3+5)=23

Some questions posed by the author:
Can a subtractive magic triangle be formed from the first 3n natural numbers for any order > 3?
Does a subtractive magic triangle retain it's magic property if each element is increased by a constant?
Does a subtractive magic triangle retain it's magic property if each element is multiplied by a constant?
What is the maximum and minimum differences obtainable for subtractive magic triangles of a given order?

[4] Sunday A. Amose, *Subtractive Magic Triangles*, Mathematics Teacher, 76 (5), 1983, pp 346-347

Balanced perimeter triangles

This pattern was suggested by Charles Trigg [5].

The digits 1 to 9 may be partitions into 3 groups, each summing to 15. We can place these groups onto the 3 sides of a triangle, but with no numbers at the vertices. Th each side will sum to 15.
There is one group consisting of 2, 2 and 5 integers, eight groups of 3, 3, and 3 integers, and 6 groups of 2, 3, and 4 integers.

I leave it as an exercise for the reader, to find the partitions for the first and last set. Also, how many different unique solutions are possible (don't count rotations and reflections)?

The eight partitions for 3 sets of 3 are elegantly represented by the 3 x 3 magic square. The main diagonals cannot be used because the share the common number 5. Tl three rows may be placed on the 3 sides of a triangle for one arrangement. The three columns may be placed on the sides of a triangle for another solution.
Any position is permitted for the numbers on a side, so there are a lot of possible solutions.

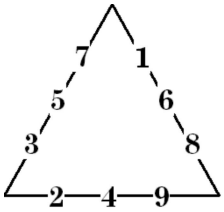


fig. 1

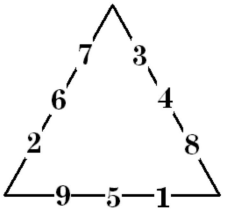


fig. 2

8	1	6
3	5	7
4	9	2

Figure 1 is the set represented by the rows of the magic square. Figure 2 is represented by the columns. While this is the simple way to determine which side to place the numbers, there are two other methods (I know of) to do so.

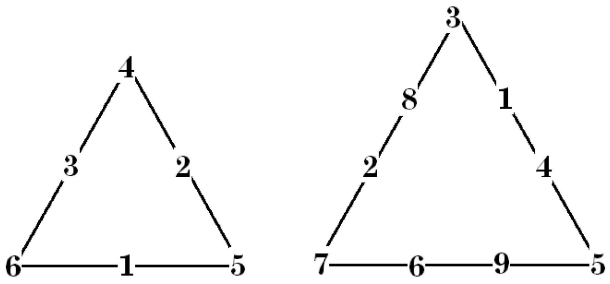
Place the numbers on the sides of the triangle in sequence, but at the end of each triad, skip one side (fig 1), or at the end of each triad, place the next number on the same side (fig 2).

For added interest, this problem may be generalized for polygons of any number of sides.

[5] Charles W. Trigg, *Triangles With Balanced Perimeters*, Journal of Recreational Mathematics, Vol. 3, No. 4, 1970, pp 255-256

Antimagic triangular arrays.

This digression from perimeter magic triangle, explores another article by Charles Trigg [6]. He investigates formations of numbers arranged in triangular arrays. How for order 3, these numbers do appear on the perimeter of a triangle (on higher orders, there are interior number as well).



Here is one such array were the sums of each side are in consecutive order (ie. 11, 12, 13). As a bonus, the order 2 triangles sums are 8, 9, 10. The inverted order 2, consisting of the 3 mid-side numbers, sum to 6, different but not in consecutive order with the others.

A few minutes of doodling produced the order 4 perimeter anti-magic triangle shown. The order 4 side sums are an arithmetic progression order (13, 20, 27). The three order 2 triangles also sum in arithmetic progression order (12, 15, 18). In an order 4 *perimeter* triangle there is no inverted center triangle.

In keeping with magic square terminology, these triangles should more properly be called heterotriangles because the sums are different but not in consecutive orde For more in-depth information, see my anti-magic squares page [7]

[6] Charles W. Trigg, *Special Antimagic Triangle Arrays*, Journal of Recreational Mathematics, Vol. 14, No. 4, 1981-82, pp 274-278
[7] For information on anti-magic squares see my anti-magic squares page

References

[1] Terrel Trotter, Jr., *Normal Magic Triangles of Order n*, Journal of Recreational Mathematics, Vol. 5,, No. 1, 1972, pp.28-32
[2] Terrel Trotter, Jr., *Perimeter-magic Polygons*, Journal of Recreational Mathematics, Vol. 7,, No. 1, 1974, pp.14-20
A scan of this article is at http://www.trottermath.net/simpleops/pmp.html
[3] John R. Hendricks, *The Magic Square Course*, 2nd edition, 1992, 522 pages +,
[4] Sunday A. Amose, *Subtractive Magic Triangles*, Mathematics Teacher, 76 (5), 1983, pp 346-347
[5] Charles W. Trigg, *Triangles With Balanced Perimeters*, Journal of Recreational Mathematics, Vol. 3, No. 4, 1970, pp 255-256
[6] Charles W. Trigg, *Special Antimagic Triangle Arrays*, Journal of Recreational Mathematics, Vol. 14, No. 4, 1981-82, pp 274-278
[7] For information on anti-magic squares see my anti-magic squares page

Other related references:
Yates, Daniel S., *Magic Triangles and a Teachers Discovery*, Arithmetic Teacher 23, 1976, pp 351-54.
Lowell A. Carmony, *A Minimathematical Problem: The Magic Triangles of Yates*, Mathematics Teacher, 70 (5), 1977, pp. 410-413
Janet Caldwell, *Magic Triangles*, Mathematics Teacher, 71 (1), 1978, pp. 39-42
Peggy A. House, *More Mathemagic from a Triangle*, Mathematics Teacher, 73 (3), 1980, pp. 191-195
Charles W. Trigg , *Second Order Perimeter–magic and Perimeter-antimagic Cubes,* Mathematics Magazine, 47 (3), 1974, pp95-97
Charles W. Trigg , *Edge Magic and Edge Antimagic Tetrahedrons*, Journal of Recreational Mathematics, Vol. 4, No. 4, 1971, pp 253-259
Charles W. Trigg , *Edge Magic Tetrahedrons with Rotating Triads*, Journal of Recreational Mathematics, Vol. 5, No. 1, 1972, pp 40-42
Charles W. Trigg, *Eight Digits on a Cubes Vertices*, Journal of Recreational Mathematics, Vol. 7, No. 1, 1974, pp 49-75
Charles W. Trigg, *Perimeter Antimagic tetrahedrons and Octahedrons*, Journal of Recreational Mathematics, Vol. 11, No. 2, 1978-79, pp 105-107
For information on Perimeter magic cubes see my unusual cubes page.



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Harvey Heinz harveyheinz@shaw.ca
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